Trigonometry

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Part I

Trigonometry

Chapter 1

Heights and Distances

1.0.1 Problems for Practice

1.0.1.1 Subjective Problems

Example : Consider a regular Octagon, the unequal diagonals are in the ratio
:

{ Hint: Let the centre to any vertex distance be r

$A_0A_2 = 2rSin\dfrac{2\pi}{8}$

$A_0A_3 = 2rSin\dfrac{3\pi}{8}$

$A_0A_4 = 2rSin\dfrac{4\pi}{8}$

So the ratio is $Sin\dfrac{2\pi}{8} : Sin\dfrac{3\pi}{8} : Sin\dfrac{4\pi}{8}$

$= 1/\sqrt{2} : \sqrt{2+\sqrt{2}}/2 : 1$

}

Chapter 2

Trigonometric Identities

2.0.1 Problems for Practice

2.0.1.1 Subjective Problems

Q1: Prove the following Identities

a) $\dfrac{2\cos 2^n\theta + 1}{2\cos\theta + 1} = (2\cos\theta - 1)(2\cos 2\theta - 1)\left(2\cos 2^2\theta - 1\right)\ldots\ldots\ldots\left(2\cos 2^{n-1}\theta - 1\right)$

2.0.1.2 Single Answer MCQ's

Q1: If $x = r\sin\alpha\sin\beta\cos\gamma$, $y = r\sin\alpha\sin\beta\sin\gamma$, $z = r\sin\alpha\cos\beta$, $w = r\cos\alpha$ then $x^2 + y^2 + z^2 + w^2$ is independent of

A) r, α, β, γ

B) r, α, β

C) α, β, γ

D) None of these

Q2: If $\tan\theta = \dfrac{\sqrt{6}a}{2} - \dfrac{1}{2\sqrt{6}a}$, then $\sec\theta - \tan\theta$ is equal to

(A) $\sqrt{6}a, \dfrac{1}{\sqrt{6}a}$

(B) $-\sqrt{6}a, \dfrac{1}{\sqrt{6}a}$

(C) $\sqrt{6}a, -\dfrac{1}{\sqrt{6}a}$

(D) None of these

Q1: $\tan\dfrac{\pi}{96} + 2\tan\dfrac{\pi}{48} + 4\tan\dfrac{\pi}{24} + 8 + 8\sqrt{3}$ is equal to

A) $\cot\dfrac{\pi}{96} - 6$

B) $\cot\dfrac{\pi}{96}$

C) 32

D) None of these

2.0.1.3 Multiple Answer Type Questions

Chapter 3

Trigonometric Equations

3.0.0.1 Single Answer MCQ's

Q1: The number of solutions of $8^{cosec^2 x} + 8^{-\cot^2 x} = \dfrac{33}{2}$, $0 \leq x \leq 2\pi$ is

A) 4

B) 6

C) 8

D) None of these

Q1: The number of distinct real solutions of

$\left(x^2+3\right)\left(cosecx+\cot x\right) - 3x + \sqrt{3}\left[x\left(cosecx+\cot x\right) - \left(x^2+3\right)\right] = 0$ on $-2\pi \leq x \leq 2\pi$ is

A) 1

B) 2

C) 4

D) None of these

Q3 : If $-\pi \leq \theta \leq \pi$ and r, θ satisfy $r\sin\theta = 3$ and $r = 3\left(2+3\sin\theta\right)$, then the number of ordered pairs (r, θ), which are solutions is/are

A) 0

B) 1

C) 2

D) None of these

Q4: If $0 \leq x, y, z, t \leq 2\pi$ and $\sin x + \dfrac{\sin y}{2} + \dfrac{\sin z}{3} = -t^2 + 2t - \dfrac{17}{6}$, then the value of $x + y + z$ is

A) $\dfrac{3\pi}{2}$

B) $\dfrac{5\pi}{2}$

C) $\frac{7\pi}{2}$

D) $\frac{9\pi}{2}$

3.0.0.2 Multiple Answer Type Questions

Q1: If $\cos\theta = a$ for exactly one value of $\theta \epsilon \left[0, \frac{7\pi}{6}\right]$, then the value of 'a' can be

A) 0

B) $\frac{\sqrt{3}}{2}$

C) -1

D) $-\frac{1}{2}$

Q2: If $\sin^9 x + \cos^6 x = 1$ in the interval $-3\pi \leq x \leq 2\pi$, then which of these statements is/are correct

A) The number of roots of the equation is 7.

B) The sum of roots is -4π

C) The ratio of the number of roots on the left side of zero to the number of roots on the right side on the right side of zero, on the number line is $\frac{4}{3}$.

D) None of these

Chapter 4

Properties of Triangle

4.0.0.1 Single Answer MCQ's

Q1: The angles of a right angled triangle are in A.P. The ratio of the area of circumcircle of the triangle to the area of the triangle is

A) $\frac{2\pi}{\sqrt{3}}$

B) $\frac{\pi}{\sqrt{3}}$

C) $\frac{\pi}{3}$

D) None of these

Q1: In a $\triangle ABC$, with sides a,b,c and $\angle B = \frac{\pi}{6}$, $\angle C = \frac{\pi}{4}$. The area of the triangle is

A) $\frac{(\sqrt{3}-1)}{4}a^2$

B) $\frac{1}{2}a^2$

C) $\frac{(\sqrt{3}+1)}{4}a^2$

D) None of these

Q2. The area of $\triangle ABC$ is $(b+c)^2 - a^2$. Then which of following statements is true

A) $\angle A$ is acute

B) $\angle A$ is right angle

C) $\angle A$ is obtuse

D) None of these

4.0.0.2 Multiple Answer Type Questions

4.0.0.3 Assertion-Reason Type Questions

Q1: Statement 1: If the sides of a triangle are 3,4,5 then the altitudes of the triangle are $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$.

Statement 2: If the sides of a triangle are in A.P. then the altitudes of the triangle are in H.P.

4.0.0.4 Linked Comprehension Type Questions

Comprehension 1: For a quadrilateral with sides a, b , c , d , semi-perimeter 's' and sum of any two opposite angles $= 2\alpha$, the area $\triangle$ is given by

$\triangle^2 = (s-a)(s-b)(s-c)(s-d) - abcd\cos^2\alpha$

Now, we take a special quadrilateral which can be inscribed in a circle C_1 and a circle C_2 is inscribed in it

Q1: The area of a special quadrilateral $\triangle$ is given by

A) $\sqrt{s(s-a)(s-b)(s-c)}$

B) $\frac{s^2}{3}$

C) $\sqrt{abcd}$

D) $\sqrt{3abcd}$

Q2: The radius of the inscribed circle C_2 is given by

A) $\frac{\triangle}{s}$

B) $\frac{2\triangle}{s}$

C) $\frac{3\triangle}{s}$

D) $\frac{4\triangle}{s}$

Q3: Cosine of the angle between the diagonals of the special quadrilateral is given by

A) $\pm\frac{ab-cd}{s^2}$

B) $\pm\frac{ac-bd}{\triangle}$

C) $\pm\frac{ab-cd}{ab+cd}$

D) $\pm\frac{ac-bd}{ac+bd}$

Q4: If B is the angle between sides a and b of the special quadrilateral then $tan\frac{B}{2}$ is given by

A) $\sqrt{\dfrac{ab}{cd}}$

B) $\sqrt{\dfrac{cd}{ab}}$

C) $\dfrac{ab}{cd}$

D) $\dfrac{cd}{ab}$

Chapter 5

Inverse Trigonometry

5.0.1 Problems for Practice

5.0.1.1 Single Answer MCQ's

Q1: The number of solutions of the equation $\sin^{-1}(1-x) - 2\sin^{-1}(x) = \frac{\pi}{2}$ is/are

a) One

b) Two

c) Three

d) None of these

Q2: If $\frac{1}{\sqrt{2}} < x < 1$, the number of solutions of the equations

$\tan^{-1}\left(\frac{1}{x} - 1\right) + \tan^{-1}\left(\frac{1}{x}\right) + \tan^{-1}\left(\frac{1}{x} + 1\right) = \tan^{-1}(3x)$ is

A) 0

B) 3

C) 4

D) None of these

Q3: The equation $\cos^{-1} x = 3\cos^{-1} a$ has a solution for

A) all real values of a

B) all real values of a satisfying $a \leq 1$

C) all real values of a satisfying $\frac{1}{2} \leq a \leq 1$

D) all real values of a satisfying $\frac{1}{\sqrt{2}} \leq a \leq \frac{\sqrt{3}}{2}$

5.0.1.2 Multiple Answer Type Questions

5.0.1.3 Linked Comprehension Type Problems

Comprehension 1: A function $f:\left(\frac{5\pi}{2},\frac{7\pi}{2}\right)\to(-1,1)$ is defined by $f\left(x\right)=\sin x$. Its inverse is denoted by $f^{-1}\left(x\right)=\sin^{-1}x$. Another function $g:(2\pi,3\pi)\to(-1,1)$ is defined by $g\left(x\right)=\cos x$. Its inverse is denoted by $g^{-1}\left(x\right)=\cos^{-1}x$. An onto function $h:(-1,0)\to R_h$ is given by $h\left(x\right)=f^{-1}(x)-g^{-1}(x)$. Now answer the following questions.

Q1: $g^{-1}(-x)$ is given by

a) $\pi-g^{-1}\left(x\right)$

b) $5\pi-g^{-1}(x)$

c) $\frac{5\pi}{2}-g^{-1}(x)$

d) None of these

Q2: Set 'R_h' is

a) $\left\{\frac{\pi}{2}\right\}$

b) $\left(\frac{5\pi}{2},3\pi\right)$

c) $\left(0,\frac{\pi}{2}\right)$

d) None of these

Q3: If a function $v:(-\infty,0]\to\left[3\pi,\frac{7\pi}{2}\right)$ is defined by $v\left(x\right)=f^{-1}\left(\frac{2^x-2^{-x}}{2^x+2^{-x}}\right)$, then $v(x)$ is

a) Injective Only

b) Surjective Only

c) Bijective

d) None of these.

Part II

Problem Set

Chapter 6

JEE Mains Problems

Q1: If $sin\left(\frac{\pi}{18}\right) sin\left(\frac{5\pi}{18}\right) sin\left(\frac{7\pi}{18}\right) = K$, then the value of $sin\left(\frac{10K\pi}{3}\right)$ is :

[JEE Mains 2nd April 2026 Shift 1 Q11]

(A) $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(B) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2}$

{ Solution : https://www.youtube.com/watch?v=qj-bKAbdLpE}

Q2: Let $S = \{x\epsilon[-\pi, \pi] : sinx(sinx + cosx) = a, a\epsilon Z\}$.Then n(S) is equal to :

[JEE Mains 2nd April 2026 Shift 1 Q12]

(A) 3

(B) 6

(C) 7

(D) 9

{ Solution : https://www.youtube.com/watch?v=DZmcnFZbvK4}

Chapter 7

JEE Advanced Problems

Q1 : Let $\alpha = \frac{1}{sin60^o sin61^o} + \frac{1}{sin62^o sin63^o} + ... + \frac{1}{sin118^o sin119^o}$.

Then the value of $\left(\frac{cosec1^o}{\alpha}\right)^2$ is ______.

[JEE Advanced 2025 Paper 2 Q15]

{Solution : https://www.youtube.com/watch?v=vM-HVpY-HD4 }